

The empirical recurrence for OEIS A282438: recovering a conjecture that is not written down

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Abstract

OEIS A282438 records “Empirical recurrence of order 88”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 217$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 88, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 88 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282438 is “Number of nX5 0..1 arrays with no 1 equal to more than three of its king-move neighbors, with the exception of exactly one element.”. It has offset 1 and begins

0, 58, 6498, 152726, 3997136, 98841792, 2301995900, 52857163712, ...

The entry states:

Empirical recurrence of order 88 (see link above)

As of the “Last modified” line on the live entry (Feb 15 2017 10:23:32, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1682$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 217$ of the 1682, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 217$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 434$ exact terms of the model returns order 88 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	28	c_{23}	−2168200877833660	c_{45}	185967287804235544124	c_{67}	33127998508678
c_2	−62	c_{24}	−19435977939762182	c_{46}	182510409940337867318	c_{68}	12685007833155
c_3	290	c_{25}	−19030982168489250	c_{47}	−193769892270874758386	c_{69}	42914720574149
c_4	−39665	c_{26}	−21802552722258409	c_{48}	−48302765957282328018	c_{70}	−10202183308870
c_5	−56164	c_{27}	109190926813199640	c_{49}	−141108259478444211278	c_{71}	−5951001638056
c_6	−1423225	c_{28}	262859383997945961	c_{50}	425292845543741124193	c_{72}	57271623476169
c_7	7883574	c_{29}	441985098008536910	c_{51}	509768706134994899074	c_{73}	29172696184063
c_8	−14961492	c_{30}	−57133812264376415	c_{52}	55123813143981676807	c_{74}	−2756366680753
c_9	403409948	c_{31}	−970865399876183816	c_{53}	−376820183762937079036	c_{75}	11981611626336
c_{10}	−721653595	c_{32}	−1699449393802874989	c_{54}	−462812054973159787997	c_{76}	−315389439229
c_{11}	7198168898	c_{33}	−1371420232846465610	c_{55}	−101459069217469840826	c_{77}	−2130177525140
c_{12}	−33805150404	c_{34}	−1546392455234327831	c_{56}	312188541672479828638	c_{78}	8268092285691
c_{13}	69611418340	c_{35}	−730532215669239870	c_{57}	375262423469962133754	c_{79}	−58482091313
c_{14}	−682697492802	c_{36}	8373254626440187378	c_{58}	70822640596954859604	c_{80}	−717074679253
c_{15}	900140314504	c_{37}	20959817302865658466	c_{59}	−241325297797634780872	c_{81}	197259368018
c_{16}	−6367407956271	c_{38}	13184954763560951260	c_{60}	−217184297718006621466	c_{82}	704261311170
c_{17}	18342043929830	c_{39}	−27377279547930875658	c_{61}	10279939536152141480	c_{83}	−14060207107
c_{18}	−21800121668062	c_{40}	−44652630533317475238	c_{62}	105054396145113545689	c_{84}	80317686417
c_{19}	216977316897898	c_{41}	−11165639737830301660	c_{63}	68373339311887900068	c_{85}	48009550734
c_{20}	−40324235418828	c_{42}	10848959729505575855	c_{64}	18436667989018280756	c_{86}	−1433449515
c_{21}	976241412231378	c_{43}	−28477098543655119460	c_{65}	−35928064538876364966	c_{87}	−844105154
c_{22}	−1913675481558514	c_{44}	15716810951501287007	c_{66}	−26059306137033437656	c_{88}	−422812057

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 88, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $88 + 4$ terms removed returns order 88 as well, so $\deg q_{\min} = 88 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $88 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 88 that this sequence satisfies — and that family turns out to have exactly one member.

References

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